

POWER BI WEEK 5

Introduction

This Week 5 Power BI project focuses on healthcare data preparation and data modelling. The healthcare dataset contains patient details, medical conditions, admission information, hospital details, insurance provider, billing amount, medication and test results. The data is cleaned and transformed to improve consistency and prepare it for analysis. Separate Dim_patient, Dim_Admission, and Billing tables are created and connected to form a structured data model for further dashboard analysis.

Name: Durga devi S

Department: B.Sc Data Science and AI – 2nd
Year

Roll No: E25DA018

Figure 1: Healthcare dataset imported into Power Query for data preparation and transformation.

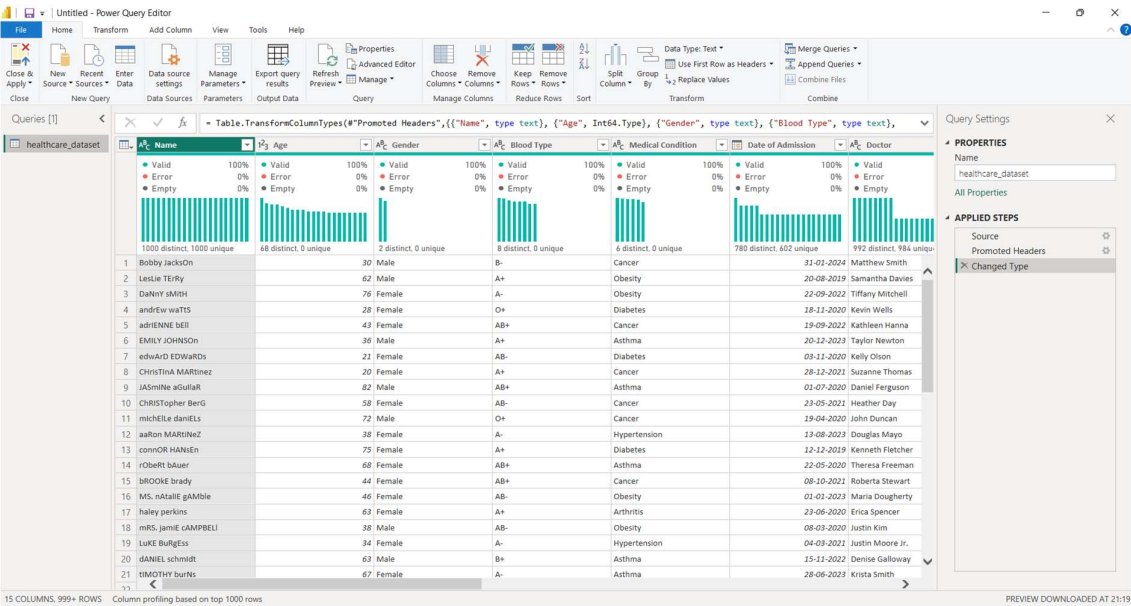


Figure 2: Categorical text fields cleaned using Trim and Clean transformations in Power Query.

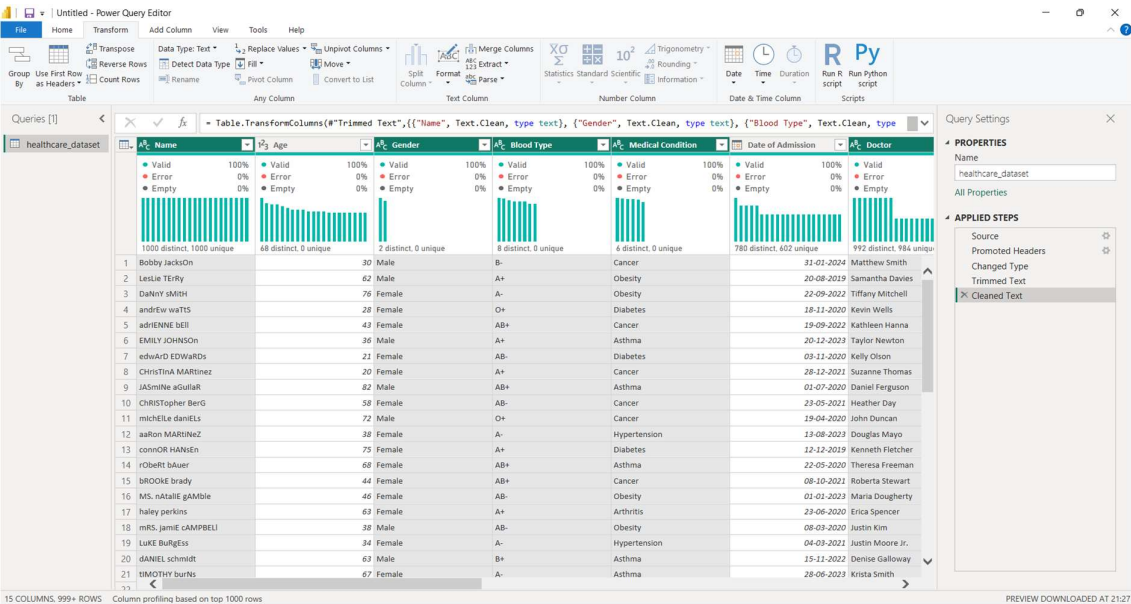


Figure 3: Name column standardized using the Capitalize Each Word transformation.

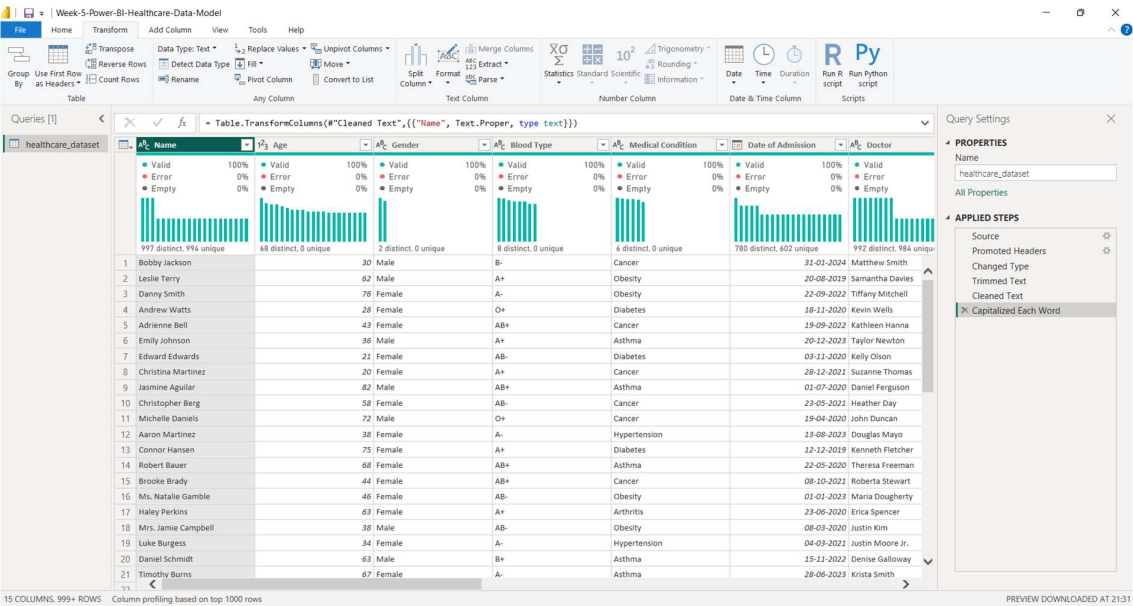


Figure 4: Age Group created as a conditional column based on patient age.

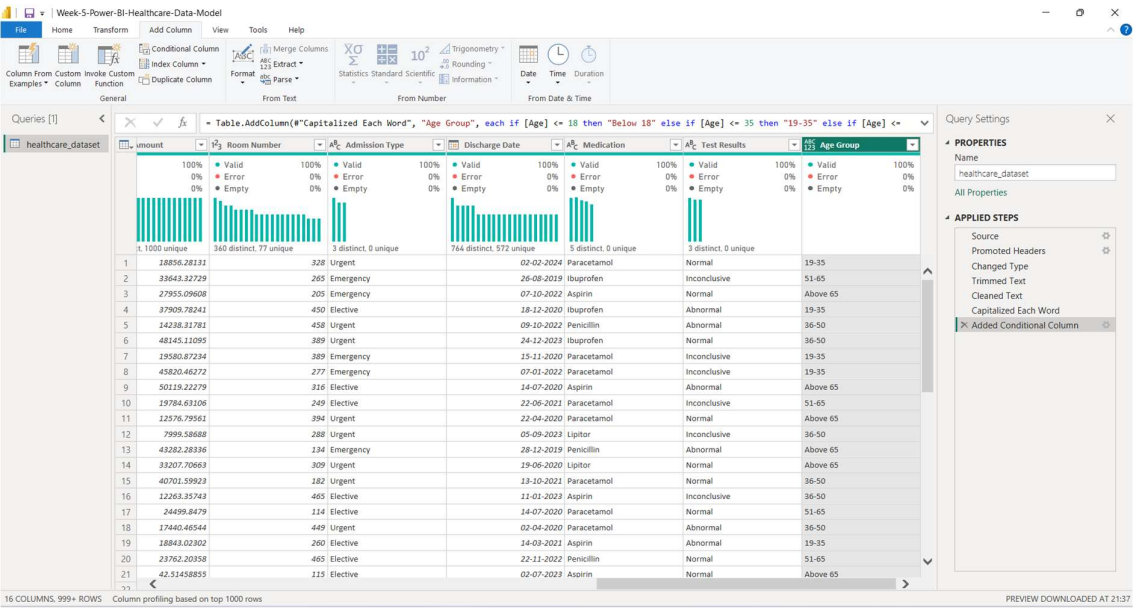


Figure 5: Billing table created with the required billing and admission-related fields.

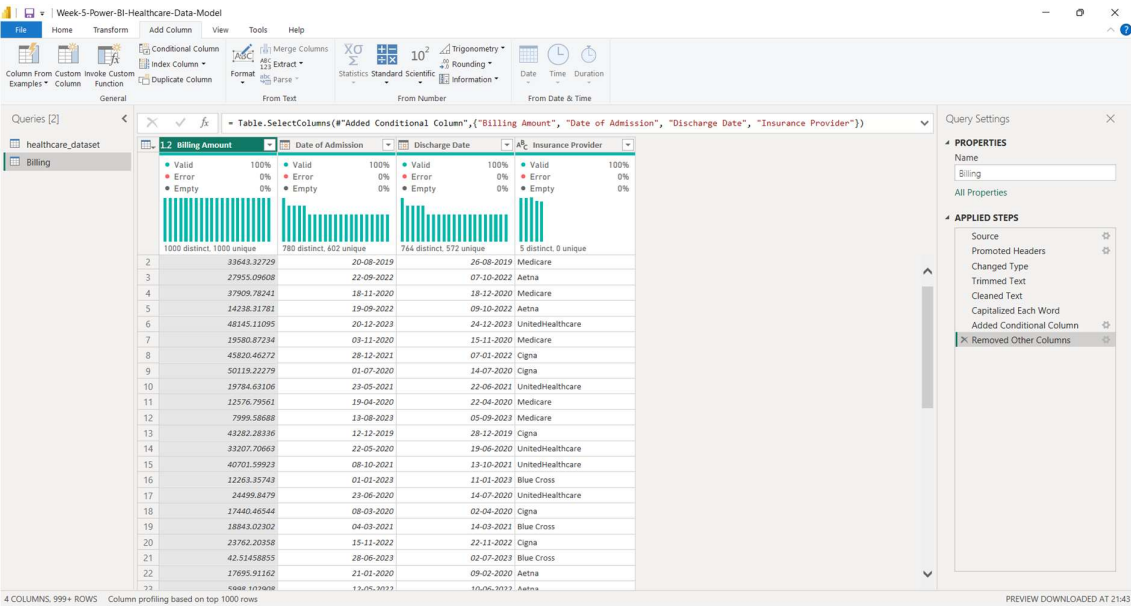


Figure 6: Dim_patient table created from the prepared healthcare dataset for patient-level modelling.

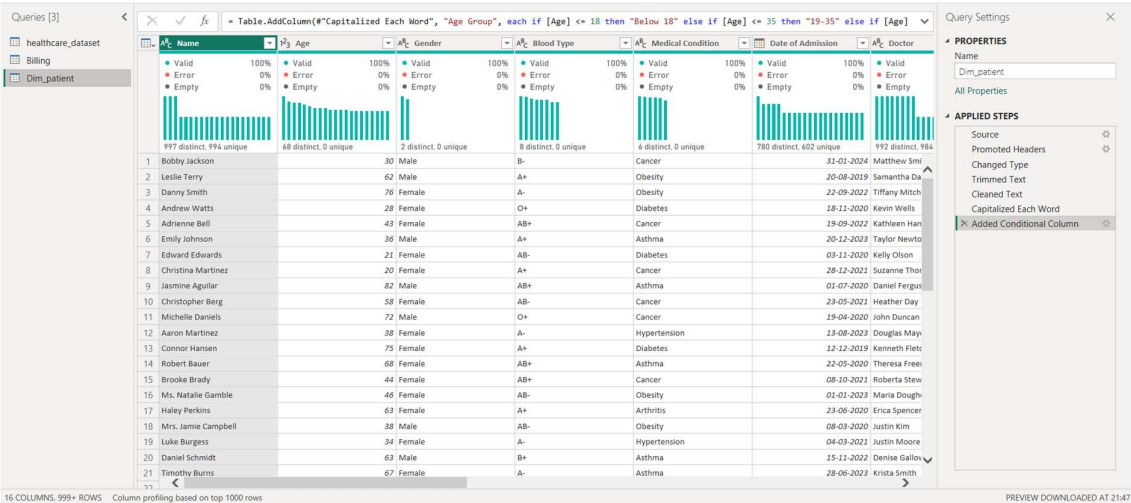


Figure 7: Patient ID created using an index column in the Dim_patient table.

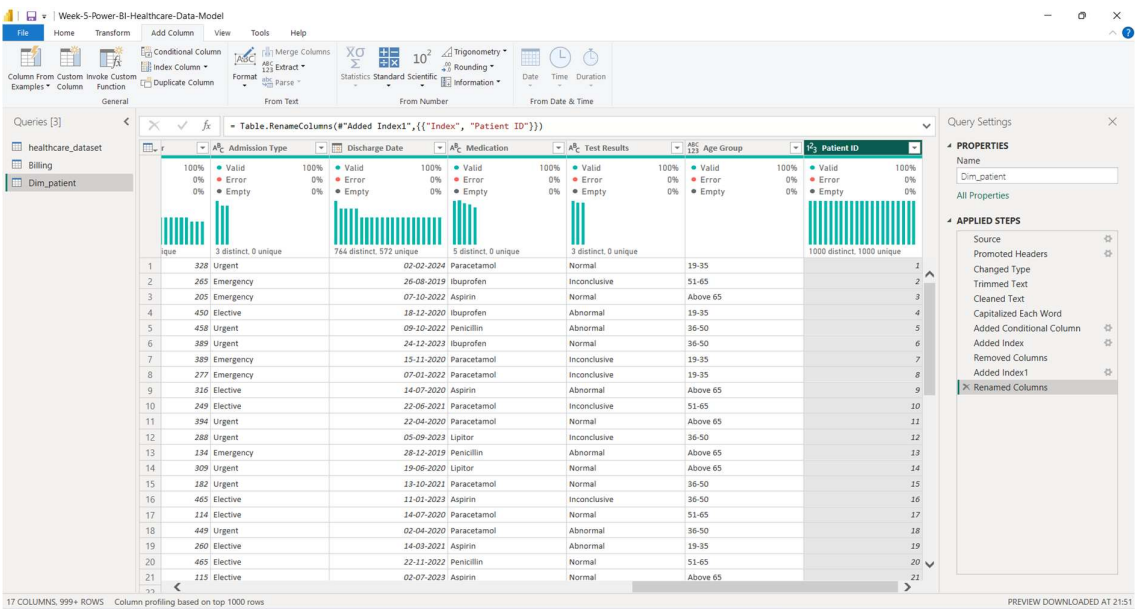


Figure 8: Billing table prepared with patient information for merging with the Dim_patient table.

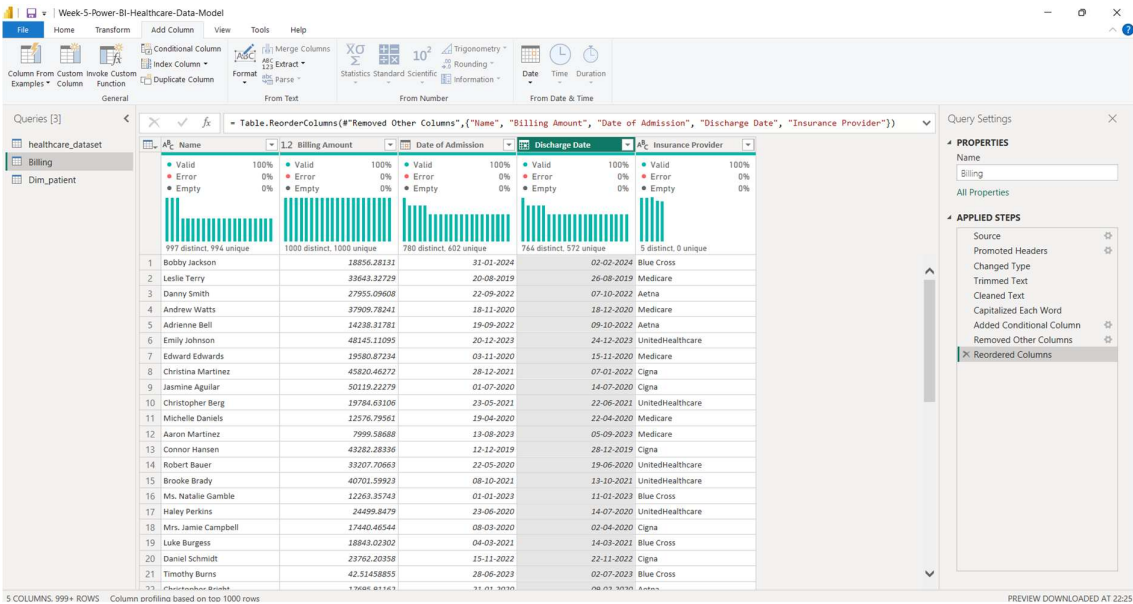


Figure 9: Billing table merged with Dim_patient using a Left Outer join based on the Name column.

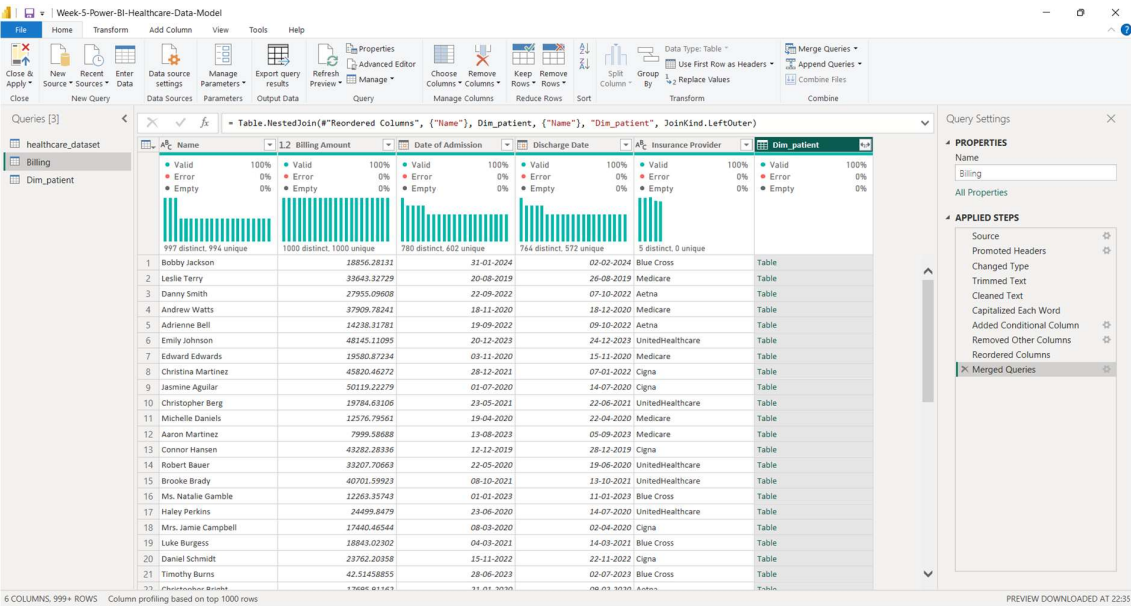


Figure 10: Patient ID expanded from the merged Dim_patient table into the Billing table.

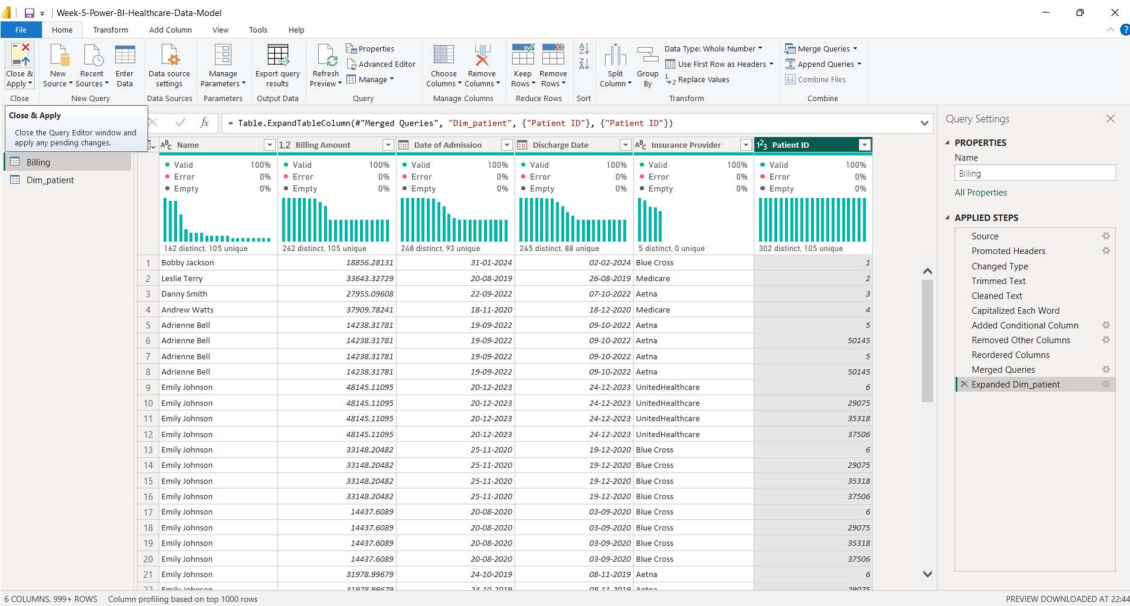


Figure 11: Final Billing table after removing the temporary Name column and retaining Patient ID.

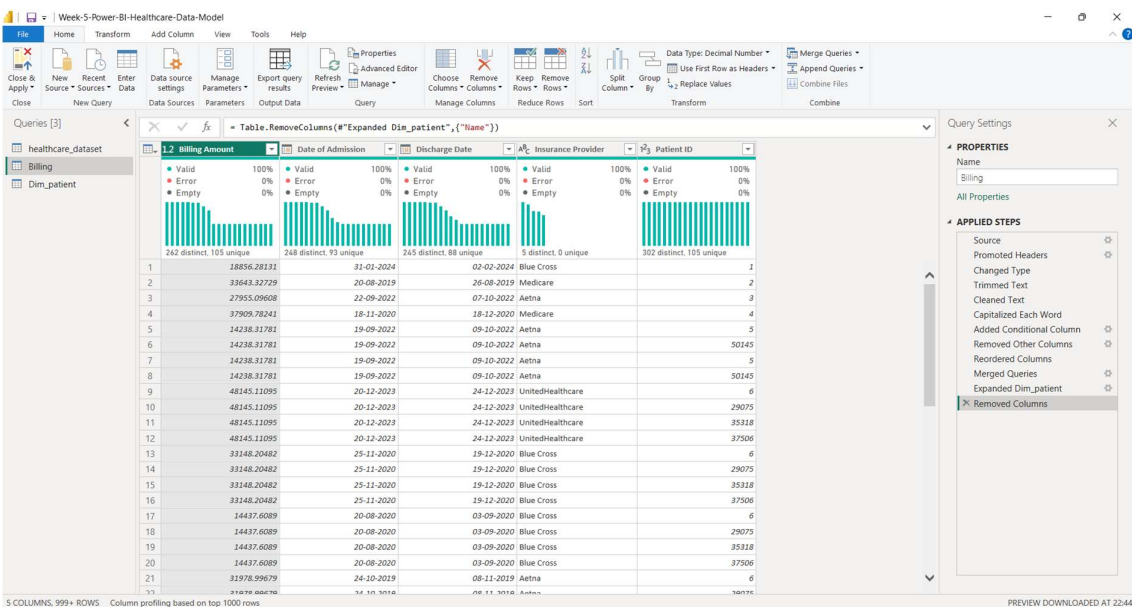


Figure 12: Billing table merged with the healthcare_dataset using the Name column and a Left Outer join.

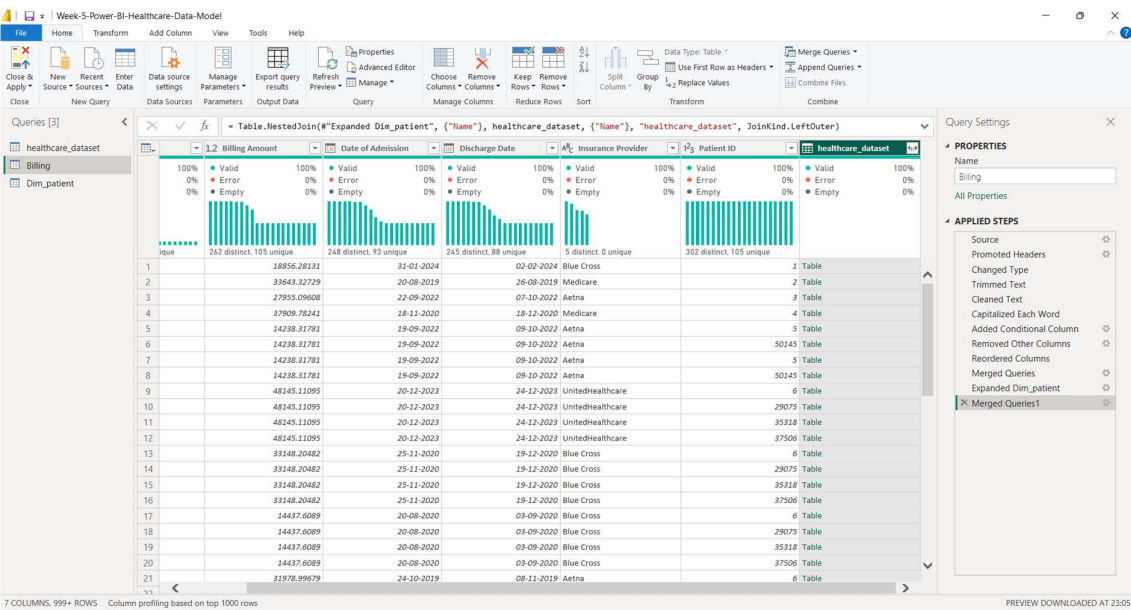


Figure 13: Healthcare details expanded from the merged healthcare_dataset into the Billing table.

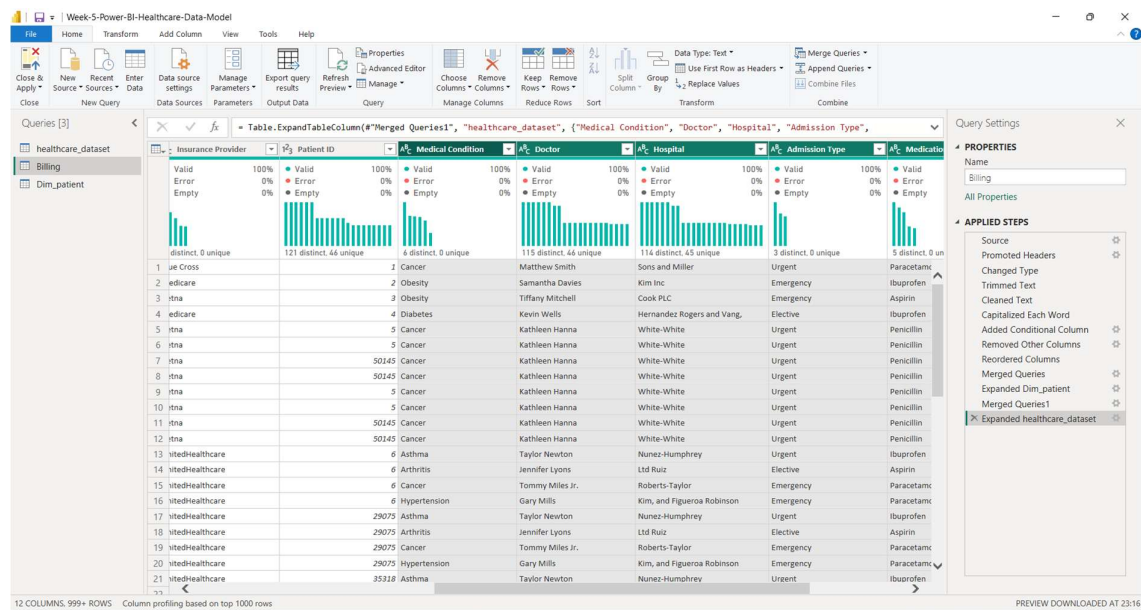


Figure 14: Transformed Billing data successfully loaded into the Power BI data model.

Week-5-Power-BI-Healthcare-Data-Model - Last saved: Today at 11:19 PM

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Structure Relationships

Data

Billing Amount	Date of Admission	Discharge Date	Insurance Provider	Name	Patient ID	Medical Condition	Doctor	Hospital	Admission Type	Medication	Test Results
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	2022	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	2022	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	2984	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	2984	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	6087	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	6087	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	7340	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	7340	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	7432	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	7432	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	7689	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	7689	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	11432	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	11432	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	14895	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	14895	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	15188	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	15188	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	20431	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	20431	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	24295	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	24295	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	24680	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	24680	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	29773	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	29773	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus
37039.965351956	10 January 2023	29 January 2023	Medicare	Michael Williams	30811	Hypertension	Kyle Ward	Brown-Bell	Emergency	Penicillin	Inconclus

Table: Billing (4,62,170 rows)

Search

Billing

Dim_patient

healthcare_dataset

Figure 15: Dim_Admission table created with required columns and Admission ID.

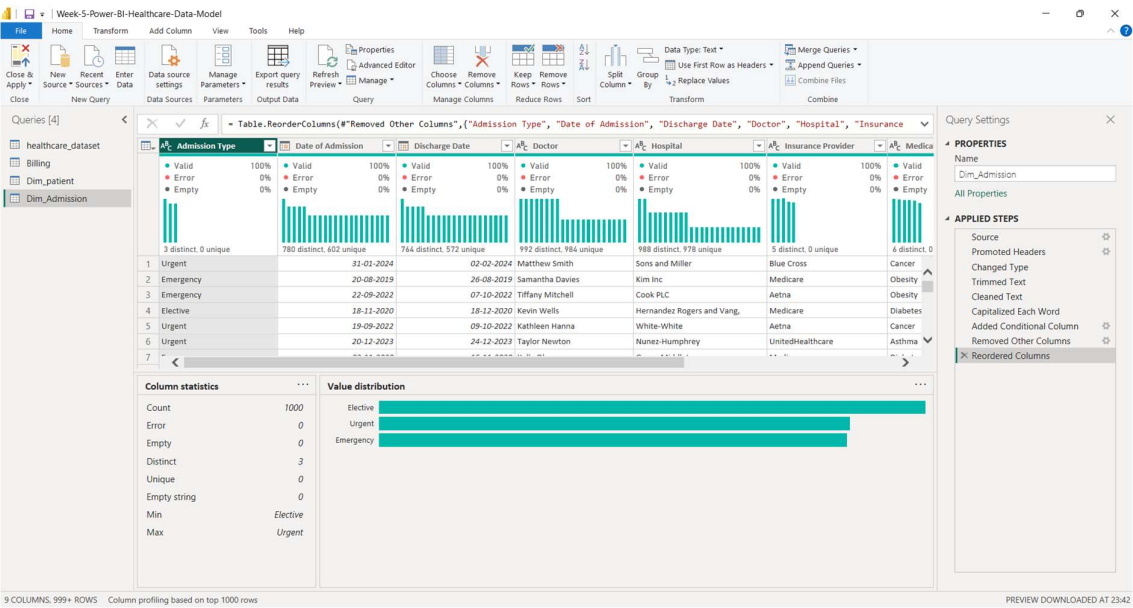


Figure 16: Billing table merged with Dim_Admission using a Left Outer join.

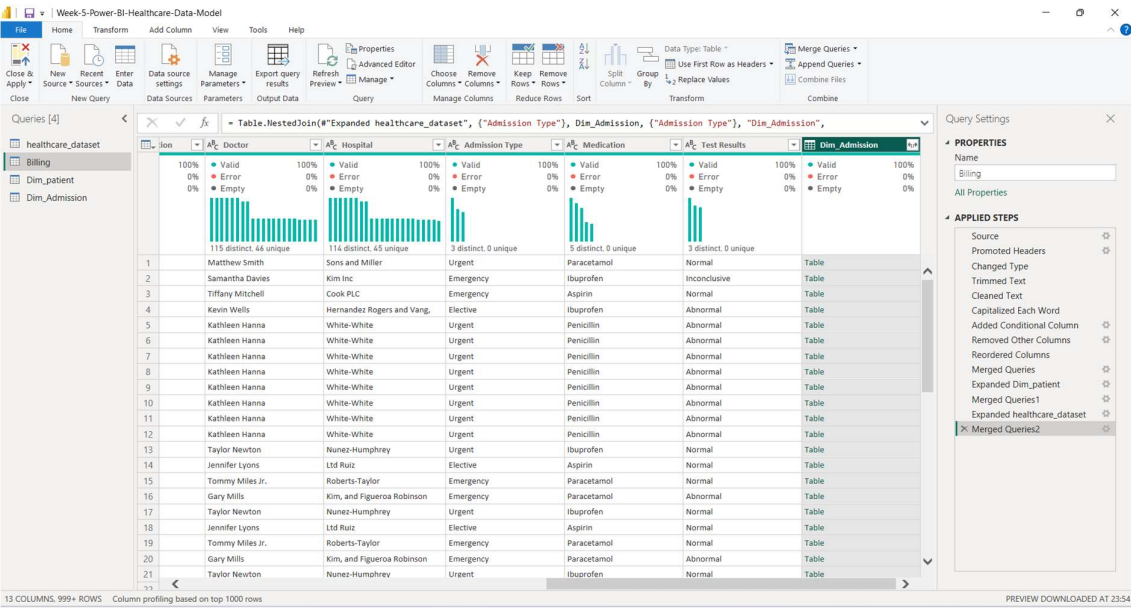


Figure 17: Admission ID expanded from Dim_Admission into the Billing table.

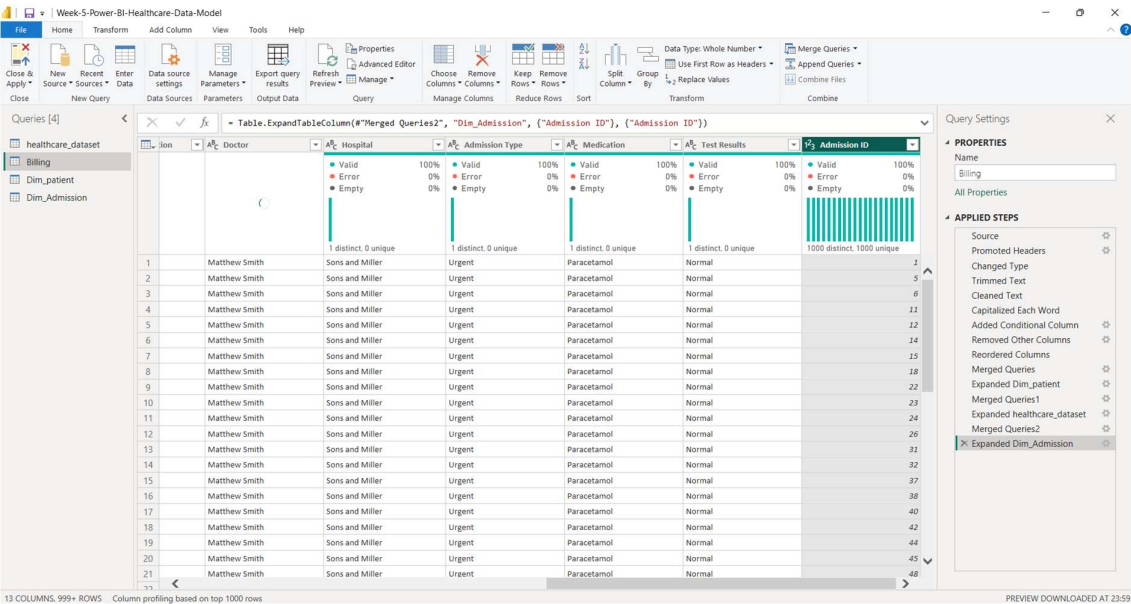


Figure 18: Final Billing table containing required billing, patient, and admission identifiers.

The screenshot displays the Microsoft Power BI Desktop interface. At the top, the title bar reads "Week 5-Power-BI-Healthcare-Data-Model". The ribbon includes tabs for File, Home, Transform, Add Column View, Tools, and Help. The main workspace shows a data model with a table named "Table.SelectColumns("Expanded Dim_Admission", "Billing Amount", "Date of Admission", "Discharge Date", "Insurance Provider", "Patient Insurance Provider", "Discharge Date", "Insurance Provider", "Insurance Provider", "Patient Insurance Provider", "Discharge Date", "Insurance Provider", "Insurance Provider", "Patient Insurance Provider")". The table has columns for Billing Amount, Date of Admission, Discharge Date, Insurance Provider, Patient Insurance Provider, Discharge Date, Insurance Provider, Insurance Provider, Patient Insurance Provider, Discharge Date, Insurance Provider, Insurance Provider, Patient Insurance Provider. The interface also shows the "Query Settings" pane on the right with "PROPERTIES" and "APPLIED STEPS" sections.

Figure 19: Final transformed tables successfully loaded into the Power BI data model.

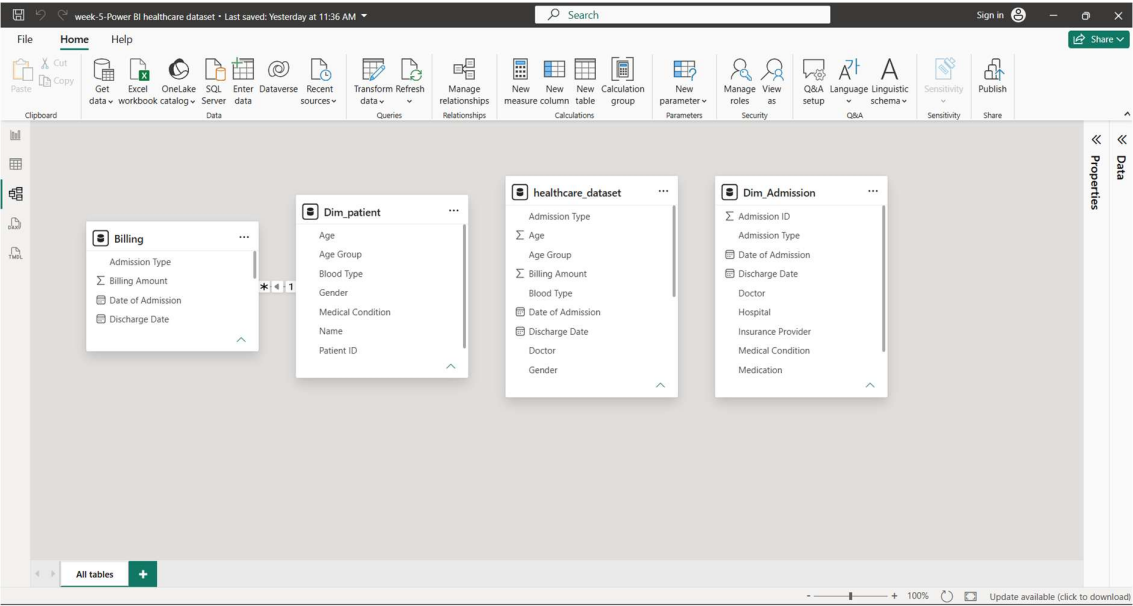
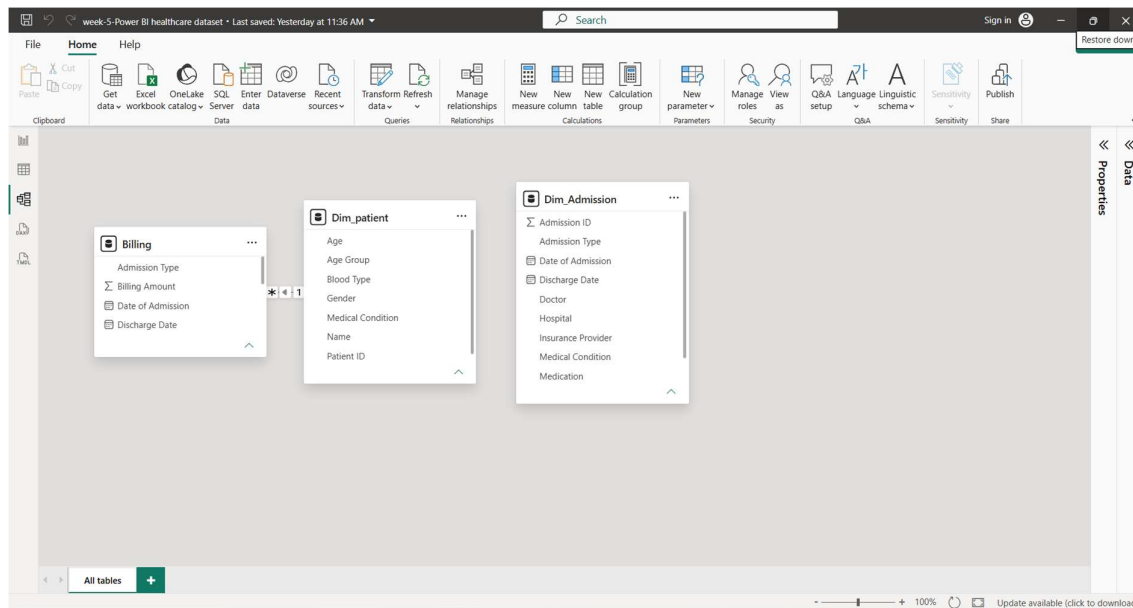


Figure 21: Original healthcare_dataset disabled from model loading, leaving the final three-table data model.



Conclusion:

In Week 5, the healthcare dataset was successfully cleaned and transformed using Power BI Power Query. Categorical fields were cleaned, names were standardized, and an Age Group column was created. Separate Billing, Dim_patient, and Dim_Admission tables were prepared, and Patient ID and Admission ID were added through merge operations. Finally, the three tables were connected using the appropriate identifiers, while the original healthcare_dataset was disabled from model loading. This created a structured data model ready for further analysis and dashboard development.